

CP k AP. 58

1) $\sin(x)$

$$U_k = (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

$$M = \frac{U_k}{U_{k-1}} = \frac{(-1)^k \frac{x^{2k+1}}{(2k+1)!}}{(-1)^{k-1} \frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} =$$

$$= \frac{(-1)(-1)^{k-1} \frac{x^{2k+1}}{(2k+1)!}}{\frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} = (-1) \cdot \frac{x^{2k+1}}{(2k+1)!} \cdot \frac{(2(k-1)+1)!}{x^{2(k-1)+1}} =$$

$$= (-1) \cdot \frac{x^{2k} \cdot x}{(2k+1)!} \cdot \frac{(2(k-1)+1)!}{x^{2k} \cdot x^{-2} \cdot x} =$$

$$= (-1) \cdot \frac{x^2 \cdot (2(k-1)+1)!}{(2k+1)!} \cdot \frac{(2k-1)!}{(2k+1)!} =$$

$$= - \frac{x^2 \cdot (2k-1)!}{(2k+1)!} = - \frac{x^2 \cdot (2k-1)!}{(2k-1)! \cdot (2k) \cdot (2k+1)} =$$

$$= - \frac{x^2}{4k^2 + 2k}$$

$$M = - \frac{x^2}{4k^2 + 2k}$$

2) $\cos(x)$

$$u_k = (-1)^k \frac{x^{2k}}{(2k)!}$$

$$M = \frac{u_k}{u_{k-1}} = \frac{(-1)^k \frac{x^{2k}}{(2k)!}}{(-1)^{k-1} \frac{x^{2(k-1)}}{(2(k-1))!}} = \frac{(-1)(-1)^{k-1} \frac{x^{2k}}{(2k)!}}{(-1)^{k-1} \frac{x^{2(k-1)}}{(2(k-1))!}} =$$

$$= - \frac{x^{2k}}{(2k)!} \cdot \frac{(2k-1)!}{x^{2k-2}} = - \frac{x^{2k}}{(2k)!} \cdot \frac{(2k-2)!}{x^{2k} \cdot x^{-2}} =$$

$$= - \frac{(2k-2)! \cdot x^2}{(2k)(2k-1)(2k-2)!} = - \frac{x^2}{4k^2 - 2k}$$

$$M = - \frac{x^2}{4k^2 - 2k}$$